

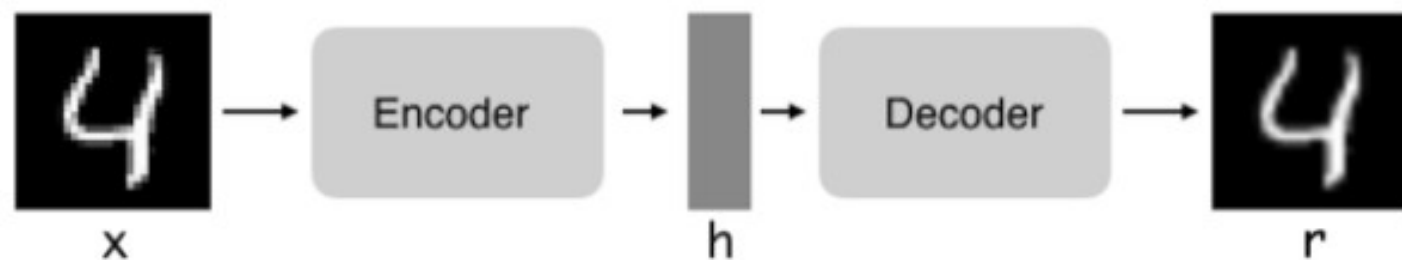


Module 4

Deep Learning: AutoEncoders

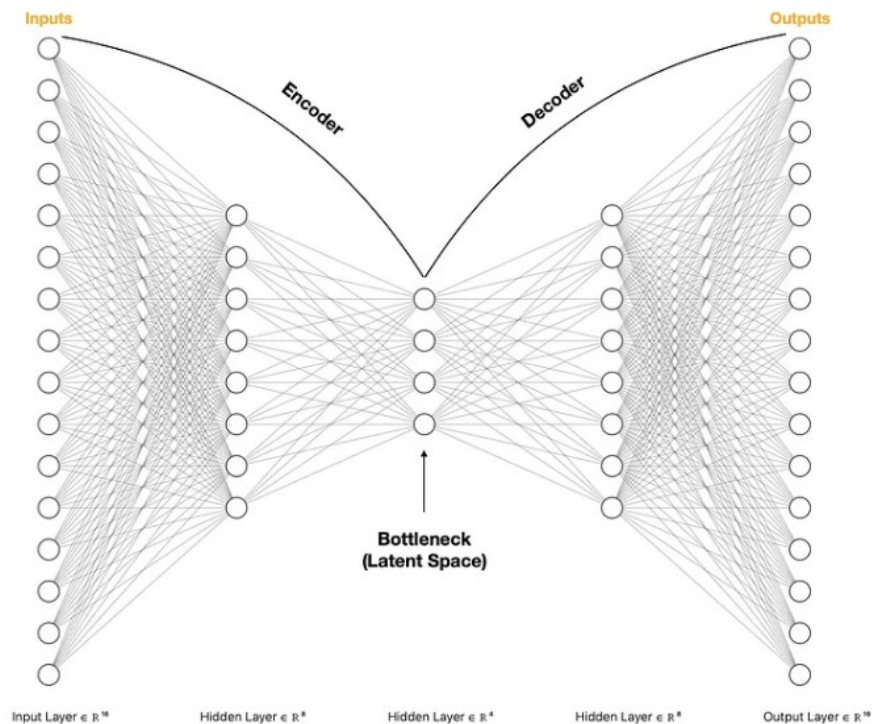


DL: AutoEncoders



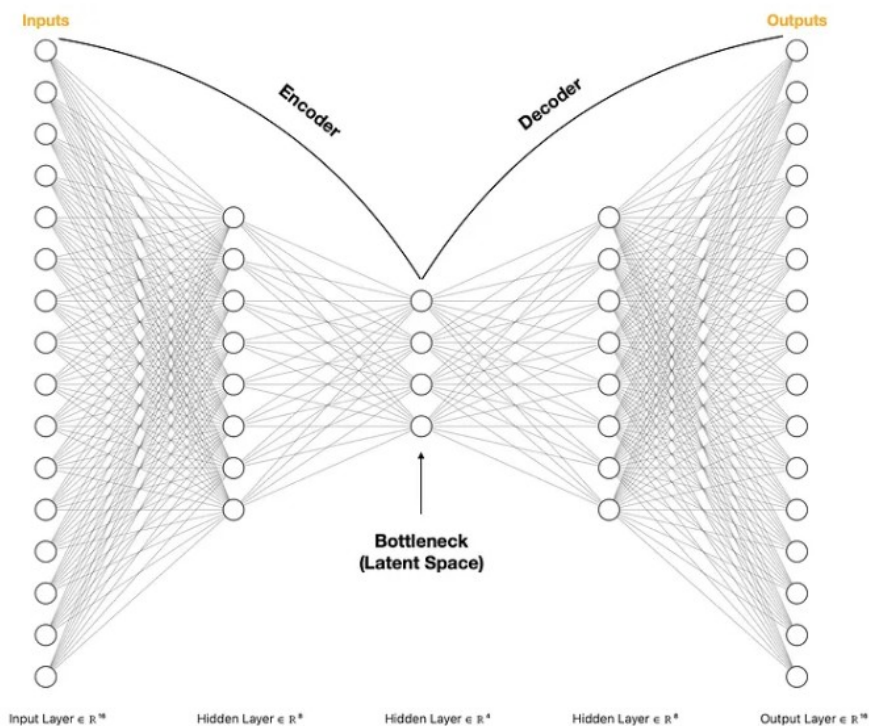
AutoEncoders

- ❑ Unsupervised learning NN architecture used to learn data codings
- ❑ Trained to reconstruct input data after compression
- ❑ Learns two functions: Encoder and Decoder



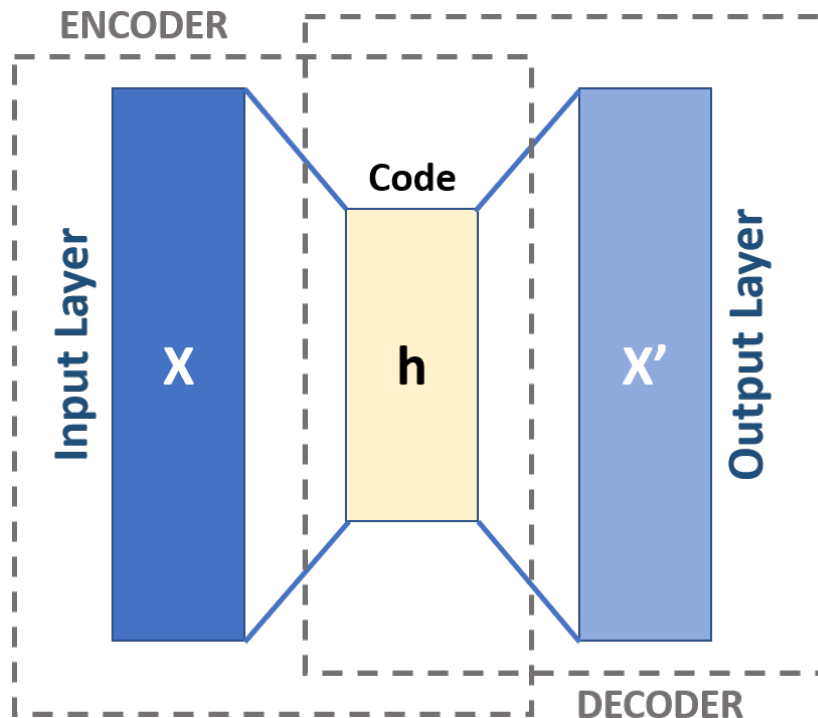
AutoEncoders

- ❑ Encoder: transforms the input data to a lower dimension
- ❑ Decoder: recreates the input data.



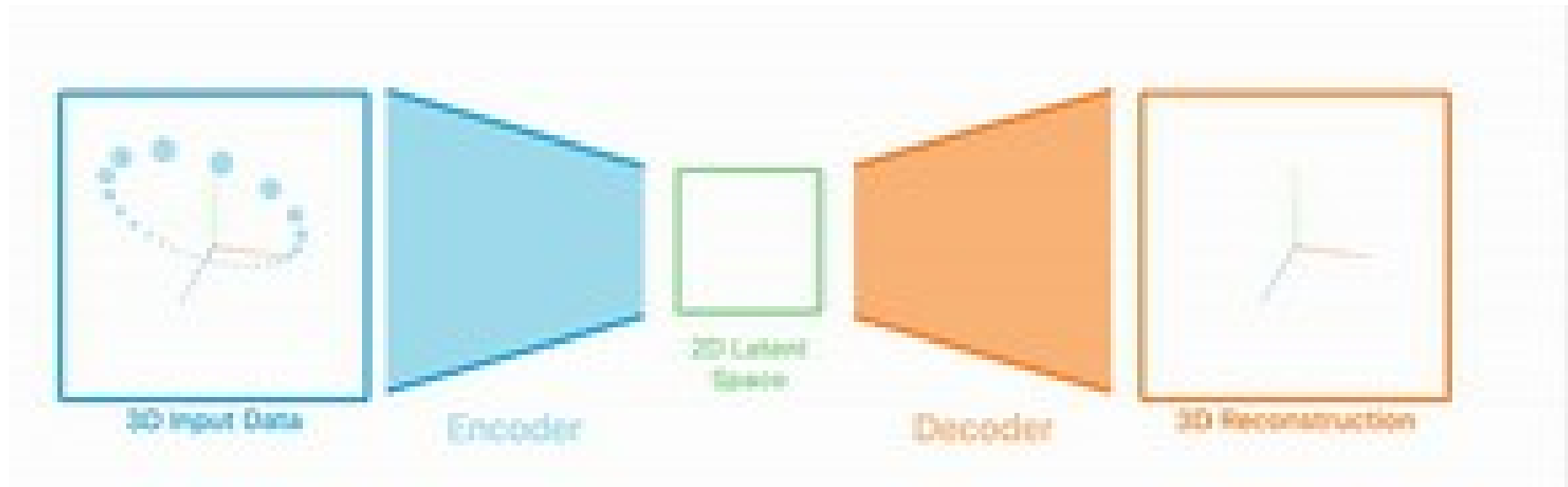
Schema of an AutoEncoder

- ❑ Input and Output layers have the same neurons (input vars)
- ❑ Encoder maps the message to a code (latent space or bottleneck)
- ❑ Decoder reconstructs the code



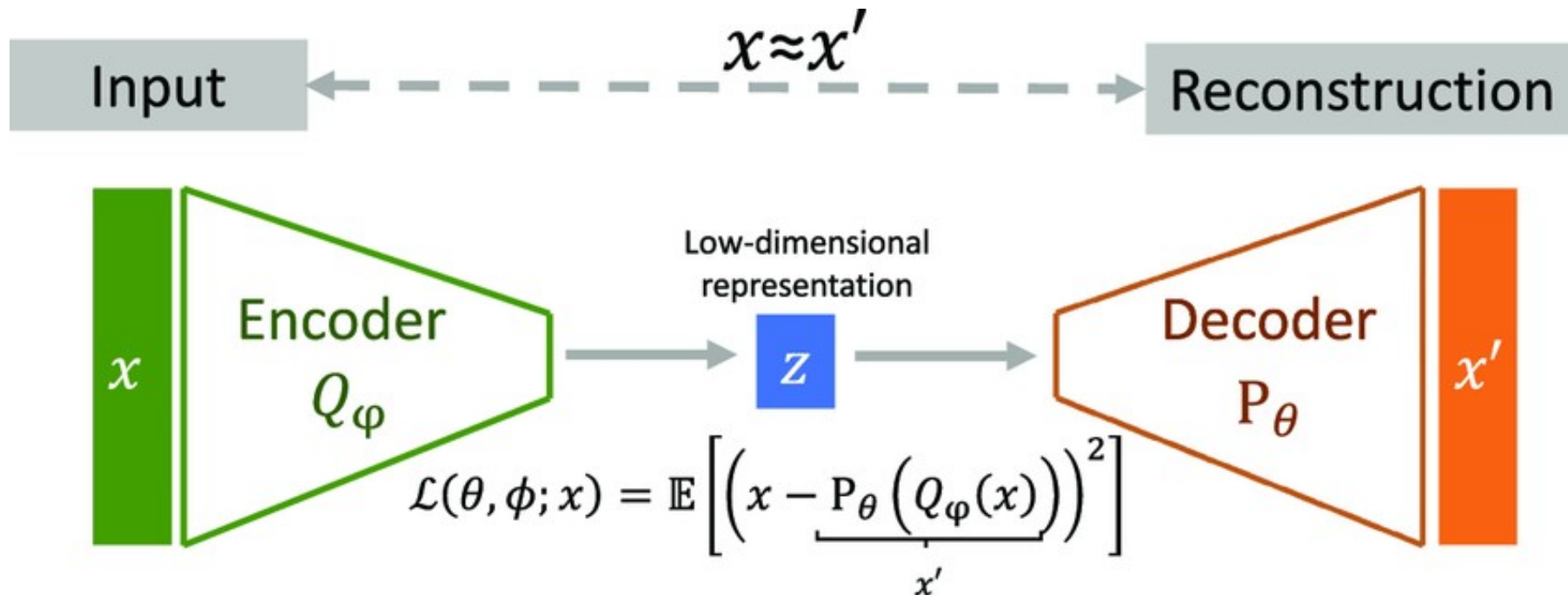
Layers

- ❑ Between encoder and decoder, the layer(s) act as a bottleneck
- ❑ Compressed knowledge representation of the original input
- ❑ The compressed data is also called the latent space



AutoEncoders

- ❑ Goal of autoencoder is to recreate the input values
- ❑ The reconstruction loss measures how good the autoencoder is

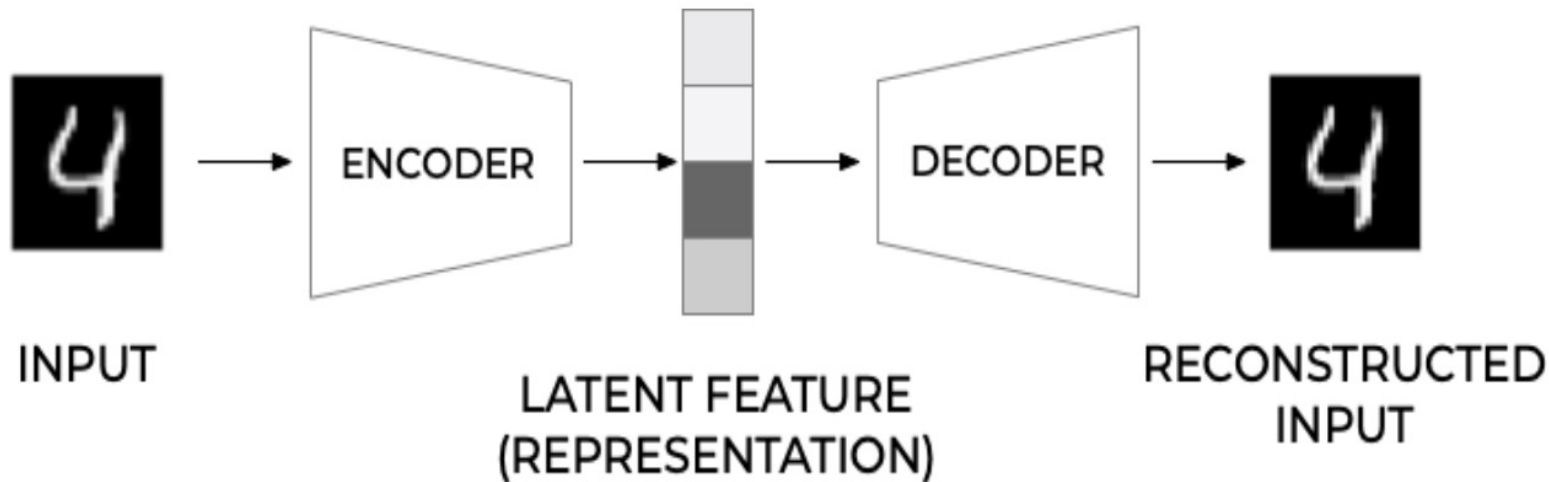


Why use AutoEncoders?

- ❑ Dimension reduction
- ❑ Noise reduction (denoising)
- ❑ Anomaly detection
- ❑ Feature extraction

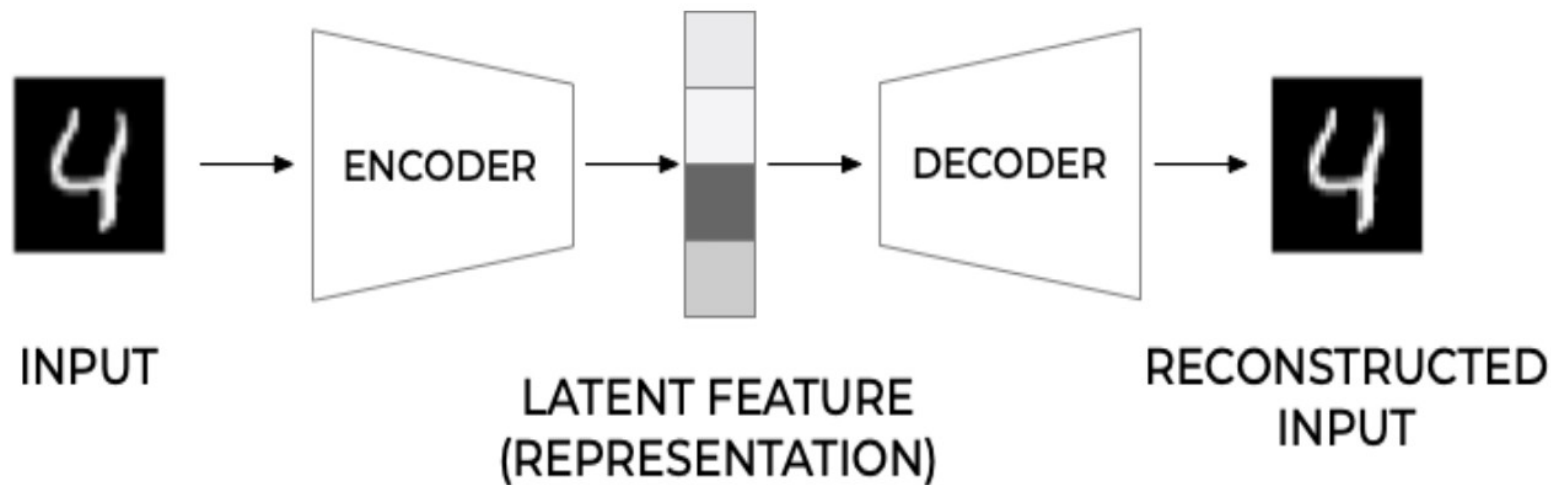
Mathematical Formulation

- Encoder: $h = g(x)$
- Decoder: f
- Reconstruction: $\tilde{x} = f(h)$
- Loss: Minimize Cost Function (MSE or Binary Cross-Entropy)



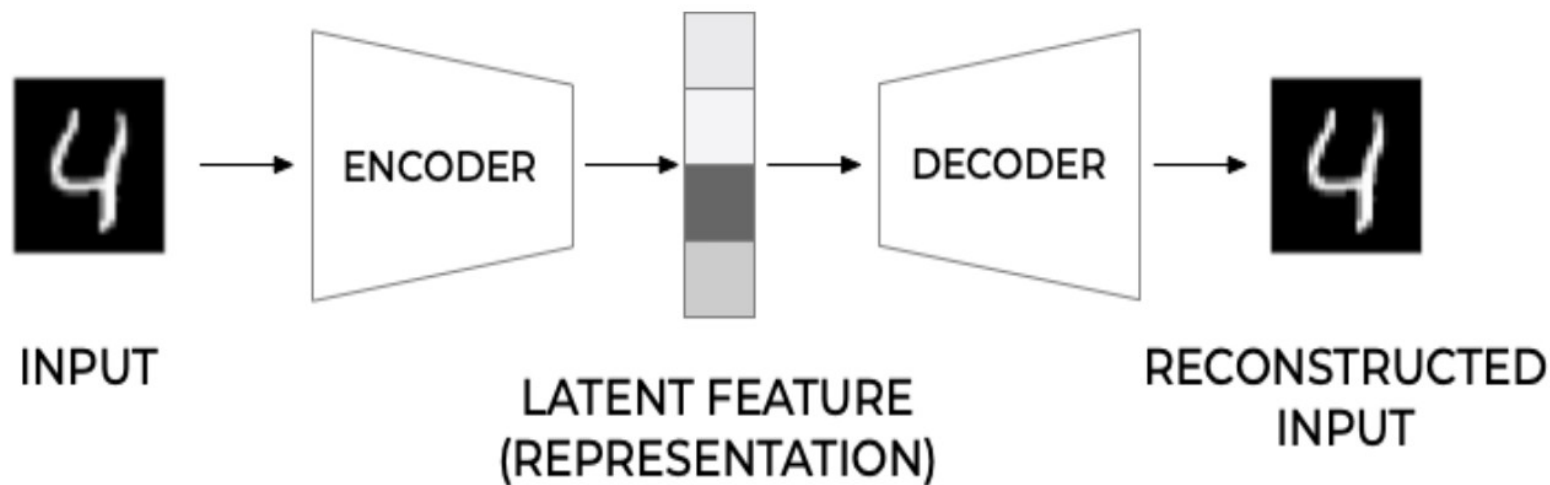
Dimensionality Reduction

- ❑ Bottleneck: Latent space has lower dimension than input
- ❑ Latent space also captures key features
- ❑ Autoencoders outperform PCA on large or nonlinear data



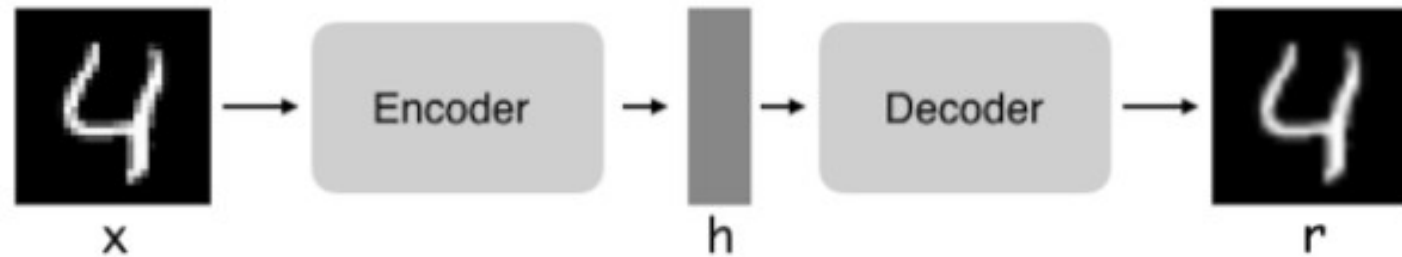
Reconstruction Error

- ▣ Difference between the original input x and reconstructed $\tilde{x}$.
- ▣ The observation MSE reconstruction error is:
- ▣ $\text{MSE}_i = \frac{1}{k} \|x - \tilde{x}\|_2^2$



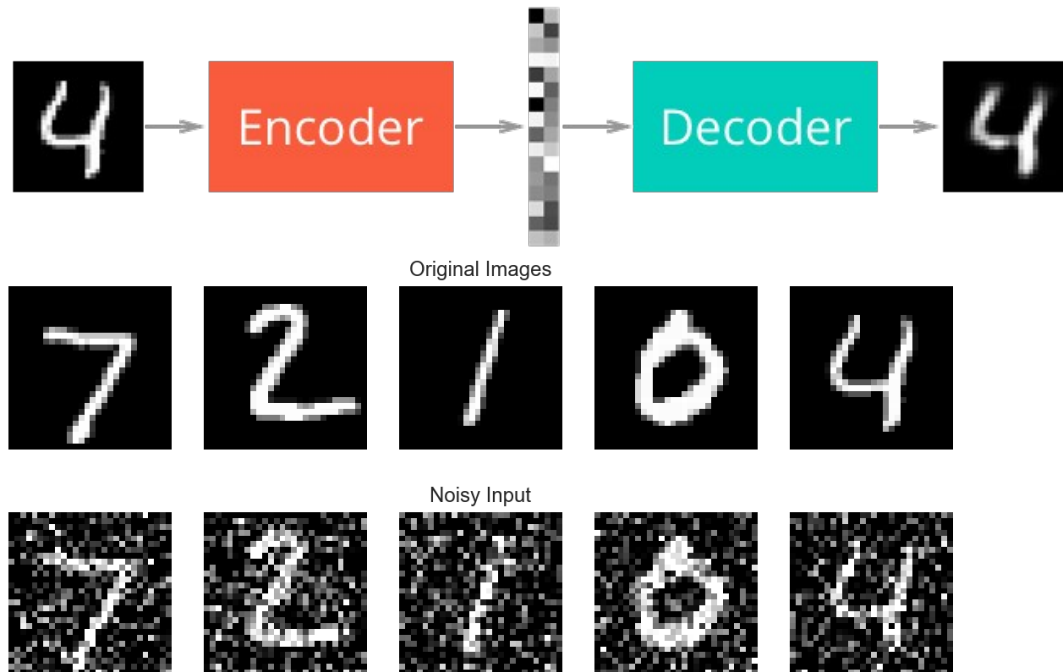


Types of AutoEncoders



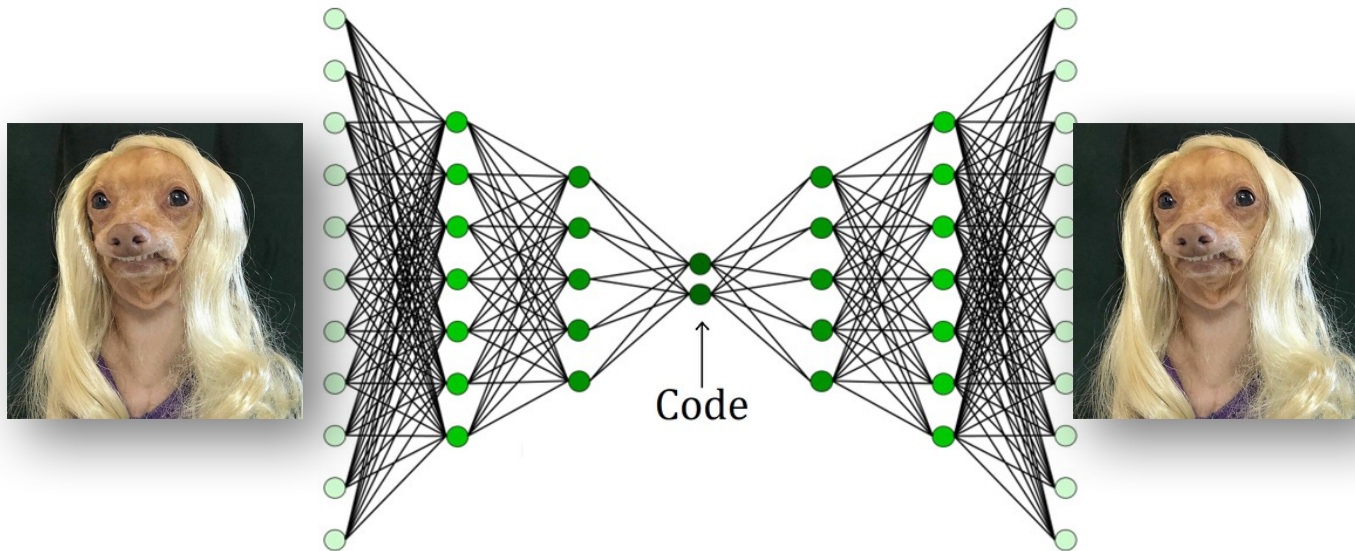
Types of AutoEncoders

- ❑ There are many types of autoencoders, we will focus on:
 - ❑ Undercomplete autoencoders
 - ❑ Sparse autoencoders
 - ❑ Denoising autoencoders



Deep AutoEncoders (Undercomplete)

- ❑ Take an input and predicts it as output.
- ❑ Using multiple hidden layers we call these deep autoencoders.

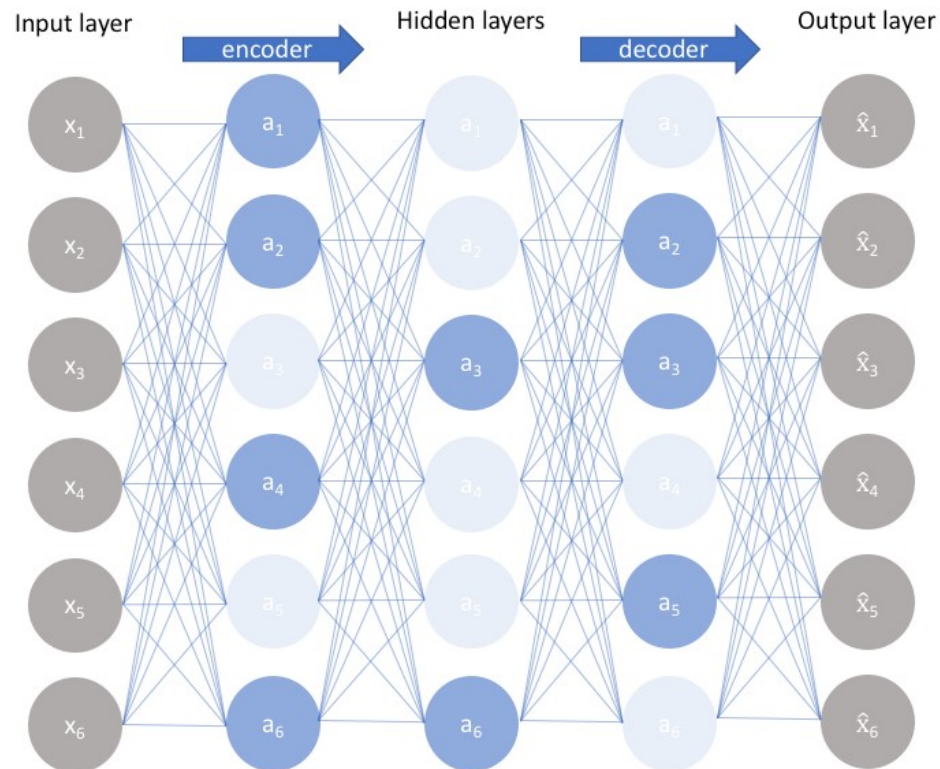




Python

Sparse AutoEncoders

- ❑ Autoencoder with a sparsity penalty
- ❑ Common sparsity penalties include L1 and L2 norms





Python

Denoising AutoEncoders

- ❑ The goal is to remove noise from a signal
- ❑ We add noise to the images but predict images without the noise.





Python